# Data 205 Final Project Report
# Historical Unemployment in the United States
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## Introduction & Goals:

For this project, I chose to analyze the unemployment rate across the US for various states. Primarily Arizona, Arkansas, Maine, New Jersey, California, and Tennessee will be my focus, although I will brush on all states. I will be focusing on historical unemployment trends, looking to uncover events and geographic differences that have impacted how unemployment has evolved.

## Tools:

This project was fully completed in R Studio, with a wide variety of different packages finding various uses along the way. Packages such as tidyr, tidyverse, and dplyr were seen and used consistently across a large portion of this project, supplementing nearly every corner of it. The package janitor was found notably in my data cleaning R Markdowns, and ggplot2 was found solely in my EDA and statistical analysis work. USABoundaries, sf, leaflet, and scales were a bundle of packages exclusive to my work in developing an interactive GIS map of the United States.

## Data:

For my primary source of data, I will be using the U.S. Bureau of Labor Statistics data on unemployment. Specifically I will be looking at their Local Area Unemployment Statistics and Labor Force Statistics including the National Unemployment Rate. This is a collection of unemployment data from across the United States of America, detailing percentage rates of unemployment across different years. It is on a monthly basis, defined by state FIPS codes. These datasets will be used for the brunt of my statistical analysis and visualization work, but will be supplemented by another census dataset on population counts. This dataset comes from the U.S. Census Bureau and is a very simple dataset on population count per year in a certain state, which is yet again denoted via state FIPS codes.

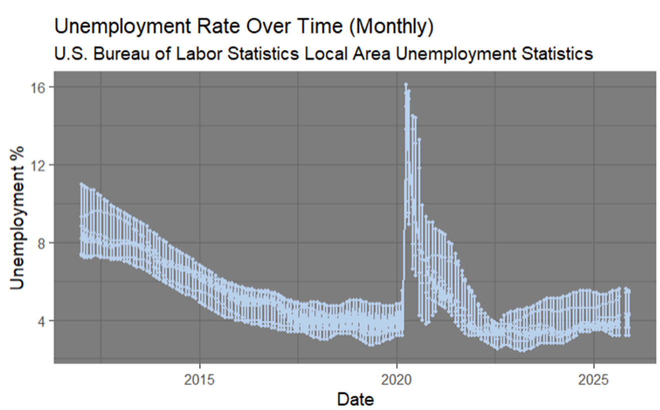
## Data Cleaning:

My primary dataset, a dataset containing my main six states from 2012-2026, was cleaned by first and foremost changing all variable names to mimic snake case (lowercase letters, spaces and hyphens replaced by underscores, no special characters). Following that, I knew I wanted state names instead of relying on FIPS codes to interpret state-by-state basis, so I created a table containing every state FIPS code and their corresponding U.S. state names. I then joined this onto my neatened dataset, ensuring that the correct numbers from variable `series_id` were being matched to the FIPS code. Finally, I ensured that all my variables were in the desired format and moved around variables for aesthetic purposes before writing the now cleaned dataset out. My supplemental dataset containing all states from 2023-2025 was cleaned in an identical manner.

My second supplemental dataset, my population dataset, was given the same snake case treatment, but had some differences when it came to everything else. First of all, it did not need a FIPS to state names change, as it already had state names. Second of all, I wanted a population density variable, so I individually defined the land area of every state and used that table to divide out population count, forming said population density variable. This dataset was also in a tall shape instead of a long one, so I took the chance to change that as well. After all of this work, this dataset was written out as well.

**Visualizations & Products:**

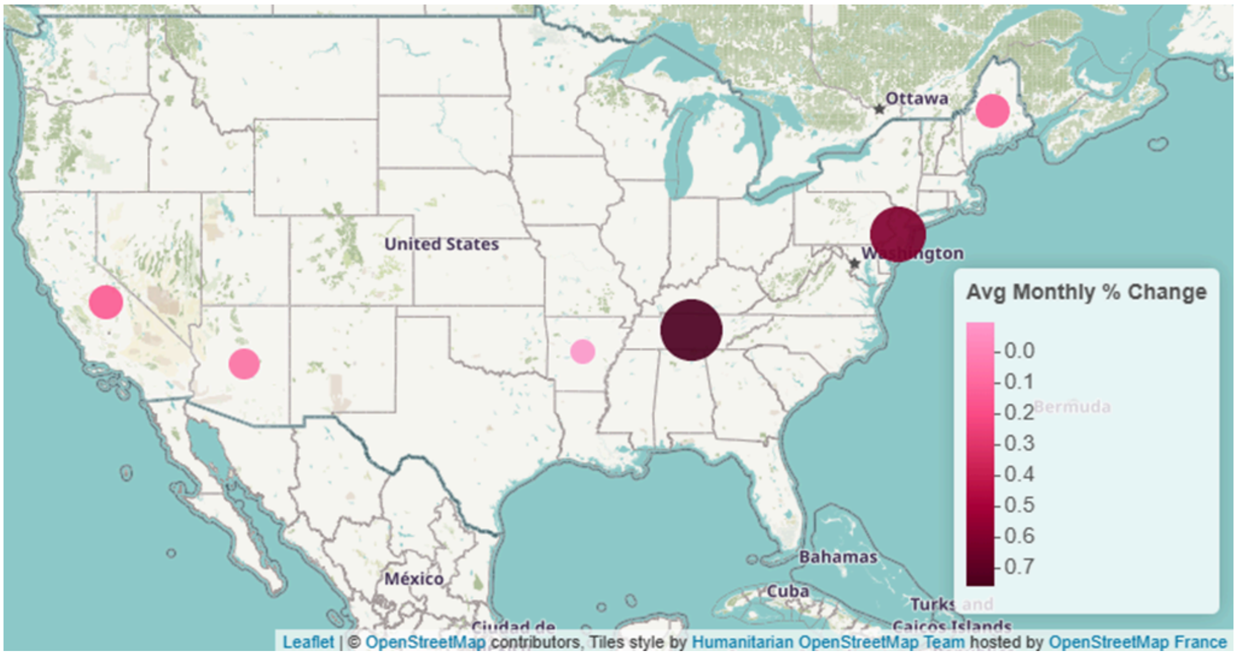
To open, a line graph of unemployment rate per month was developed, spanning the entire 2012 to 2025 time span. This graph expressed a lot in terms of historical events, notably in the 2012-2018 range, the 2018-2020 range, the 2020-2022 range, and 2023 onwards to our modern day:



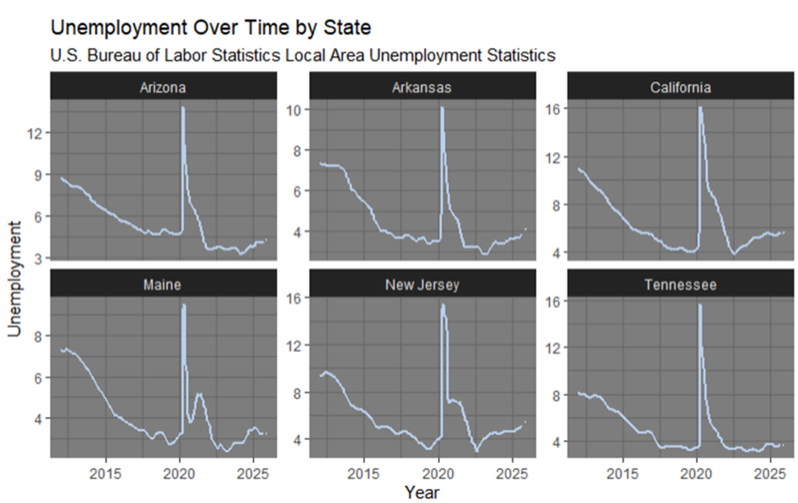

The 2012-2018 time span expresses a dropoff in the unemployment rate over time, implying a moment of time in which more people were able to get jobs. This point in time is known as the Great Recession, and was a healing point for the U.S. economy. Following that is the 2018-2020 length of time, in which the data seems to plateau out and reside lower on the graph. This point in time represents a concept known as “full employment”, where as many people as possible are able to get jobs that they need. After that is the 2020-2022 COVID-19 spike and recovery, a point in time where all economies and work forces took a massive hit as businesses shut down, people were laid off, and the world went into a global shutdown.

Fortunately, past 2023 and that entire era of time, we got through the brunt of recovery and reached back to a semblance of normal. Despite this we do begin to trend upwards slightly.

Looking past time and into geographic location, I began exploring GIS mapping:




This map confirmed a handful of my suspicions and simultaneously gave me a larger amount of questions to explore. Specifically, it did seem to be consistent with the concept of denser or smaller states having higher unemployment variation, as they tend to be typically more unstable. However, noting that Tennessee had alarmingly high unemployment rate change and Arkansas, a state that borders Tennessee, had alarmingly low unemployment rate change was quite odd. To see if I could draw up a conclusion for this, I developed a line graph of unemployment over time, but this time faceted by state:




Although this series of graphs gave me no good insight to the instabilities of the southern states, it did push forward yet another reality regarding geographic impact on unemployment. Every state graph does have a COVID-19 spike, and quite notably so. However, only some states, New Jersey and Maine, have harsh recovery spikes, implying that they have experienced complications with their recovery capabilities. This would make sense due to the small size yet dense population that these two states have, something unique to the smaller state sizes of the northeast coast states. Since the other states seem somewhat similar in their response times in comparison to the northeast states, we can assume that having good response time may also be a characteristic of southern or western states due to their sizes and more sparsely populated land.

With the constant discussion on population density, I decided to go through with a Pearson correlation test between population density and state unemployment. I first tested this with all states possible, preemptively noting that my data was right-skewed. I ended with a p-value of 0.22, so not statistically significant according to the 95% confidence interval, and a correlation coefficient of 10%, implying an incredibly weak positive correlation. Although these results are very poor, due to how skewed my data is, I did expect this. So, I logarithmically transformed my data and retested, getting a p-value near zero and a correlation coefficient of 50%. This new p-value does express statistical significance, and although the correlation coefficient isn't absolutely soaring, it is heaps better than it was before and does express a notable relationship between population density and unemployment. Out of sheer curiosity, I chose to do another Pearson correlation test, this time only including my 6 focus states. Yet again, my data was right skewed, and yet again, by flat testing the variables I did get a p-value that could not be considered statistically significant (0.09) and a correlation coefficient that was positive but low (28%). After logarithmically transforming my data and retesting it, I did get improved answers with a p-value of 0.02, showing statistical significance, and a correlation coefficient of 37.81%, expressing a weakly positive relationship. Even at a smaller scale, the correlation is consistent with other states, expressing that the effect of population density on unemployment is a consistent factor.

**Conclusion:**

In general, the goals I aimed to reach were adequately met. I found key historical moments that have contributed to how our unemployment rates have changed and I have discovered how different factors affect unemployment. Notably, I have statistically discovered how population density impacts unemployment and how geographic location creates a variety of patterns in the development of unemployment. Different clusters of states fall victim to different degrees of urbanization, work industries, population densities, and recovery speeds from crises. These subtle patterns in our economies and careers heavily change the flow of unemployment, both in incredibly quiet and incredibly loud ways.

**Experience:**

Throughout my experience with this project, the most challenging portion for me to work through was cleaning and merging my datasets. First of all, the time periods did not perfectly line up. Although my main dataset was from years 2012-2025, my population dataset would only span from 2020-2025. This would unfortunately prove to complicate some of my statistical analysis process. Second of all, The U.S. Bureau of Labor Statistics does not utilize state names as a variable, only state FIPS codes. I had to create a manual table in order to develop my own state name variable, a complication in my work that was both time consuming and challenging to face.

Every set of challenges does not come without a few diamonds in the rough, and there were quite a few rewarding high points of this project. Notably, I had the external research I conducted prior to any statistical work show itself as my project uncovered more and more information. For example, seeing an example of full employment was not something I thought I would witness throughout this project, especially given the time span I was working within.

In the future, if I were to alter the work I have done I would consider two things. First, adding demographic information. Adding details such as race, gender, political party, etc.. would be interesting, especially for statistical tests to see if variables like those change unemployment. Second, adding more states into my main data. I feel as if adding more states would've allowed me to have a deeper analysis and better grasp on geographic differences, especially if I had added a state from the northern part of the U.S..

**Next Steps:**

If I were to continue this project past what I have already achieved, there are three things I would like to do. First of all, create a forecasting model to predict the future of our unemployment rates. This would be beneficial to understanding what is relevant to unemployment, as I could look at different time periods and understand how modern day events are seemingly having an effect. Second of all, I would like to look deeper into the various degrees of unemployment that each state faces. Although global crises are relevant factors in how unemployment evolves over time, there are state-specific events and factors that may have just as much of an effect on how unemployment sways. If a new chain of restaurants opens up in Michigan, for example, many jobs will open and unemployment will decrease, but only specifically for Michigan. The third thing I would like to continue on with would be analyzing southern states and how their rates of unemployment change. Specifically, Tennessee and Arkansas are states that border one another, yet they had the highest and lowest amounts of average monthly percent change respectively. Noting this creates conflict in the potential for geographic location to affect the rate of change in unemployment, so it is generally something that stood out and I would like to understand better.

**Acknowledgements:**

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